

Hans William Alexander Hanley

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RESEARCH INTERESTS

AI safety and truthfulness; disinformation and misinformation; network analysis; natural language processing; computational social science.

EDUCATION

Ph.D., Computer Science 2025
Stanford University — Advisor: Zakir Durumeric. GPA 4.02. Thesis: *Narrative Networks: Investigating Patterns of Influence and Propaganda across International News Outlets*

M.S., Computer Science 2024
Stanford University

M.Sc., Statistical Science 2020
University of Oxford — Distinction (Highest Honours). Daniel M. Sachs Class of 1960 Graduating Scholar, Worcester College

M.Sc., Computer Science 2019
University of Oxford — Distinction (Highest Honours). Daniel M. Sachs Class of 1960 Graduating Scholar, Worcester College

B.S.E., Electrical Engineering 2018
Princeton University — Highest Honours. GPA 3.982. Concentration in Information Security and Privacy. Phi Beta Kappa; Tau Beta Pi; Sigma Xi

EXPERIENCE

Member of Technical Staff Apr. 2026–present
Microsoft AI, New York, NY

Research Scientist, Central Applied Sciences Mar. 2025–Apr. 2026
Meta, New York, NY

Health AI Student Researcher Summer 2024
Google Research, Mountain View, CA

Digital Forensic Research Lab Intern Summer 2021
Atlantic Council, Washington, DC

Data Engine Software Intern Summer 2018
Microsoft, Seattle, WA

Gmail Security Software Intern

Google, Sunnyvale, CA

Summer 2017

Google Docs and Drive Security Software Intern

Google, New York, NY

Summer 2016

NSF REU, Declarative Programming

Texas Tech University, Lubbock, TX

Summer 2015

PEER-REVIEWED PUBLICATIONS

1. **H. W. A. Hanley** and Z. Durumeric. *Hierarchical Level-Wise News Article Clustering via Multilingual Matryoshka Embeddings*. The 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025). [\[pdf\]](#)
2. **H. W. A. Hanley**, E. I. Okabe, and Z. Durumeric. *Tracking the Takes and Trajectories of English-Language News Narratives across Trustworthy and Worrisome Websites*. 34th USENIX Security Symposium (USENIX Security 2025). [\[pdf\]](#)
3. **H. W. A. Hanley** and Z. Durumeric. *Sub-Standards and Mal-Practices: Misinformation's Role in Insular, Polarized, and Toxic Interactions on Reddit*. The 28th ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2025). **Best Paper Honorable Mention**. [\[pdf\]](#)
4. **H. W. A. Hanley** and Z. Durumeric. *Twits, Toxic Tweets, and Tribal Tendencies: Trends in Politically Polarized Posts on Twitter*. The 28th ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2025). [\[pdf\]](#)
5. **H. W. A. Hanley**, Y. Lu, and J. Pan. *Across the Firewall: Foreign Media's Role in Shaping Chinese Social Media Narratives on the Russo-Ukrainian War*. Proceedings of the National Academy of Sciences (PNAS), Brief Report. [\[pdf\]](#)
6. **H. W. A. Hanley**, D. Kumar, and Z. Durumeric. *Specious Sites: Tracking the Spread and Sway of Spurious News Stories at Scale*. 45th IEEE Symposium on Security and Privacy (IEEE S&P 2024). [\[pdf\]](#)
7. **H. W. A. Hanley** and Z. Durumeric. *Machine-Made Media: Monitoring the Mobilization of Machine-Generated Articles on Misinformation and Mainstream News Websites*. 18th International AAAI Conference on Web and Social Media (ICWSM 2024). [\[pdf\]](#)
8. **H. W. A. Hanley** and Z. Durumeric. *Partial Mobilization: Tracking Multilingual Information Flows Amongst Russian Media Outlets and Telegram*. 18th International AAAI Conference on Web and Social Media (ICWSM 2024). [\[pdf\]](#)
9. **H. W. A. Hanley** and Z. Durumeric. *TATA: Stance Detection via Topic-Agnostic and Topic-Aware Embeddings*. The 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023). [\[pdf\]](#)
10. **H. W. A. Hanley**, D. Kumar, and Z. Durumeric. *A Golden Age: Conspiracy Theories' Relationship with Misinformation Outlets, News Media, and the Wider Internet*. The 26th ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2023). [\[pdf\]](#)
11. **H. W. A. Hanley**, D. Kumar, and Z. Durumeric. *Happenstance: Utilizing Semantic Search to Track Russian State Media Narratives about the Russo-Ukrainian War on Reddit*. 17th Interna-

- tional AAAI Conference on Web and Social Media (ICWSM 2023). [\[pdf\]](#)
12. **H. W. A. Hanley**, D. Kumar, and Z. Durumeric. “A Special Operation”: A Quantitative Approach to Dissecting and Comparing Different Media Ecosystems’ Coverage of the Russo-Ukrainian War. 17th International AAAI Conference on Web and Social Media (ICWSM 2023). [\[pdf\]](#)
 13. **H. W. A. Hanley**, D. Kumar, and Z. Durumeric. *No Calm in the Storm: Investigating QAnon Website Relationships*. 16th International AAAI Conference on Web and Social Media (ICWSM 2022). [\[pdf\]](#)
 14. **H. W. A. Hanley**, Y. Sun, S. Wagh, and P. Mittal. *DPSelect: A Differential Privacy Based Guard Relay Selection Algorithm for Tor*. 19th Privacy Enhancing Technologies Symposium (PoPETs 2019). [\[pdf\]](#)

PREPRINTS AND WORK IN PROGRESS

1. P. Gerard, **H. W. A. Hanley**, L. Luceri, and E. Ferrara. *Bridging the Narrative Divide: Cross-Platform Discourse Networks in Fragmented Ecosystems*. Preprint. [\[pdf\]](#)
2. **H. W. A. Hanley**, E. I. Okabe, and Z. Durumeric. *M-STANCE: Soft Contrastive Learning and Synthetic Translations for Multilingual Stance Detection*. Preprint.

DISSERTATIONS

Narrative Networks: Investigating Patterns of Influence and Propaganda across International News Outlets 2025

Stanford University, Ph.D. in Computer Science

GENerateZ: Automatic De Novo Design of Anticancer Drugs using Transcriptomic Data, Genetic Algorithms, and Variational Autoencoders 2020

University of Oxford, M.Sc. in Statistical Science

PROMISE: Provably ROBust Malware detectIon uSIng diffERential privacy 2019

University of Oxford, M.Sc. in Advanced Computer Science

DPSelect: A Differential Privacy Based Guard Relay Selection Algorithm for Tor 2018

Princeton University, B.S.E. in Electrical Engineering

SELECTED HONOURS AND AWARDS

- **Best Paper Honorable Mention**, ACM CSCW, 2025 (top 4%)
- **Rising Star in Data Science**, Stanford / UC San Diego / University of Chicago, 2024
- **AAAI Doctoral Consortium**, 2025
- **Meta Ph.D. Research Fellowship** in Computational Social Science, 2023
- **Stanford University EDGE Doctoral Fellowship**, 2020
- **National Science Foundation Graduate Research Fellowship (GRFP)**, 2020
- University of Oxford, Worcester College Prize for Academic Excellence, 2019 and 2020
- **Daniel M. Sachs Class of 1960 Graduating Scholarship**, Worcester College, University of Oxford, 2018
- **James Hayes–Edgar Palmer Prize in Engineering**, Princeton University, 2018

- **Charles Ira Young Memorial Tablet and Medal**, Princeton University Electrical Engineering, 2018
- **George B. Wood Legacy Prize**, Princeton University, 2016
- Phi Beta Kappa; Tau Beta Pi; Sigma Xi
- National Merit Scholar for Academic Excellence

SELECTED INVITED TALKS AND PRESENTATIONS

- *Narrative Networks: Investigating Patterns of Influence and Propaganda across International News Outlets* — IDEaS Center Seminar Series, Carnegie Mellon University, Pittsburgh, PA, 2026
- *Narrative Networks: Investigating Patterns of Influence and Propaganda across International News Outlets* — Google DeepMind Research Talk, London, UK, 2026
- *Narrative Networks: Investigating Patterns of Influence and Propaganda across International News Outlets* — Workshop on The Future of Social Media Research, Oxford Internet Institute, Oxford, UK, 2025
- *Narrative Networks: Investigating Patterns of Influence and Propaganda across International News Outlets* — UC Santa Cruz Security Seminar, Santa Cruz, CA, 2025
- *Tracking the Takes and Trajectories of News Narratives from Trustworthy and Worrisome Websites* — University of Cambridge Security Group Meeting, Cambridge, UK, 2024
- *Narratives of Foreign Media Ecosystems in Chinese Social Media Discussions of the Russo-Ukrainian War* — TADA: New Directions in Analyzing Text as Data, Boston, MA, 2023
- *Happenstance: Tracking Russian State Media Narratives on Reddit* — 119th American Political Science Association Annual Meeting, Los Angeles, CA, 2023
- *Influence of Foreign and Domestic Media Ecosystems on Chinese Social Media* — Politics and Computational Social Science (PaCSS), Los Angeles, CA, 2023
- *Narratives of Foreign Media Ecosystems in Chinese Social Media* — 40th Annual Meeting of the Society for Political Methodology, Stanford, CA, 2023
- *Online Information Flows and Ecosystems: Misinformation and AI-Generated Media* — Oxford Internet Institute, University of Oxford, UK, 2023
- *Online Information Flows and Ecosystems: Misinformation and AI-Generated Media* — Information Security Seminar, University College London, UK, 2023
- *Tracking the Influence of Russian State Media Narratives* — University of Florida, Gainesville, FL, 2022
- *Tracking the Influence of Russian State Media Narratives* — University of Central Florida, Orlando, FL, 2022
- *Happenstance: Tracking Russian State Media Narratives on Reddit* — Stanford Security Lunch, Stanford, CA, 2022
- *Methods in the Madness: Conspiracy Theories, Misinformation Outlets, and the Wider Internet* — Stanford Security Workshop, Stanford, CA, 2022
- *No Calm in the Storm* — Stanford Security Lunch, Stanford, CA, 2021
- *GENerateZ* — RSC-BMCS / RSC-CICAG Artificial Intelligence in Chemistry, London, UK, 2020
- *DPSelect* — Oxford Computer Science Conference, Oxford, UK, 2019
- *An Exploration of Contextualized Word Vectors for Sentiment Analysis* — Oxford Computer Science Conference, Oxford, UK, 2019

SELECTED WRITING FOR THE ATLANTIC COUNCIL DFRLAB

- Iranian accounts promote anti-Israel hashtags in the wake of the Israel-Hamas war
- New COVID-19 conspiracy theories gain traction as old ones re-emerge
- Misinformation regarding France's COVID-19 "health passes" spread on Twitter
- Iranian social media exploit US-based conspiracies regarding USS Liberty incident (with A. Kann)
- China weaponizes discovery of graves at Canadian residential schools to avoid Xinjiang criticism
- Pro-Kremlin media echo Chinese outlets that COVID lab leak theory is a U.S. disinfo campaign
- Nigerian Twitter users flock to VPNs amid dubious legal threats (with J. Le Roux)

SERVICE AND LEADERSHIP

Program committees

- AAAI 2027 (2026–2027); AISI 2027 (2026–2027)
- AAAI 2026 (2025–2026); AISI 2026 (2025–2026); IEEE S&P 2026 (2025–2026)
- AAAI-AISI 2025 (2024–2025); AAAI 2025; ICWSM 2025; ICWSM 2022

Reviewing

- IC2S2 2025; NLP4Democracy 2025; ACL 2025; ACM CSCW 2025
- IC2S2 2024; ACM CSCW 2024; ICWSM 2023
- ACM CSCW 2023 (2022); ACM CSCW 2022
- Stanford Journal of Science, Technology, and Society (2022)

Mentoring and leadership

- Stanford Computer Science Graduate Mentor (2021–2022)
- Stanford Center for Teaching and Learning, Engineering Learning Consultant (2020–2022)
- Daniel M. Sachs Global Scholarship Interviewer (2021)
- Stanford Undergraduate Research Association (SURA) Mentor (2020)
- President, National Society of Black Engineers, Princeton University Chapter

TECHNICAL SKILLS AND LANGUAGES

- **Programming:** Python, Java, C, C++, C#, Go, SQL, R, MATLAB, Git
- **ML / LLM:** PyTorch, TensorFlow, Hugging Face; transformers; LLM fine-tuning; multilingual modelling; interpretability and feature attribution; contrastive learning; LLM-based evaluation; safety and robustness analysis
- **Languages:** English (native), Mandarin Chinese (HSK 5)